

energy transition is being driven around solar power. However, other technologies like green hydrogen or carbon capture may also be crucial but they need to achieve commercial viability first. “A multi-faceted approach with short-term, medium-term, and long-term planning and policymaking is necessary for India to manage the transition in the coming decades,” Lolla says.

According to the WEF, building strong industries and supply chains at home constitutes a central point of India’s strategy in advanced solutions. This, combined with a rapidly-expanding market, positions India to become an important hub for clean hydrogen. “India also plans to produce 5 million metric tonnes of

ment needed across these solutions is unprecedented.

For example, achieving a path to net-zero emissions by 2030 will need carbon capture and storage (CCS) to scale to 20 times the current capacity, energy storage to 35 times, clean hydrogen production to 70 times, and sustainable aviation fuel to 190 times, he says. Additionally, mass deployment of new advanced modular nuclear reactors will also be needed. Combined annual investment is likely to exceed \$500 billion in the next 10 years.

“There is a huge market that needs to be served and there is room for many players. India, with its cheap renewable energy [potential], skilled workforce, and vibrant entre-

emphasises innovative procurement, offtake agreements, and research and development investment to bolster these sectors,” he adds.

Advanced technology has its challenges, and the creation of viable business cases for investment remains a major hurdle.

“Energy system benefits (like low emissions) need to become clear. But it is also critical to know how many jobs these investments can bring, what could be the healthcare savings and how industrial competitiveness can be improved. When these benefits are clear, the economic case for deployment of advanced energy solutions is established. And then, policymakers and regulators can introduce mechanisms to redistrib-

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1 Kolaczowski, on his trip to India, met domestic industry players in advanced energy solutions

2 He says India has set bold goals, focusing on clean hydrogen, energy storage, carbon capture, green aviation fuel

3 India, with its cheap renewable energy potential is poised to become a major player in green energy

4 India also has the potential to provide alternative options in advanced energy solutions to the world

clean hydrogen by 2030. To do this, the country wants to create 40GW of electrolyser manufacturing capacity, which will more than double the world’s total capacity as of 2023,” adds Kolaczowski.

Can India compete globally in advanced energy solutions? Currently, China offers advanced technology at a cheaper rate, but countries are looking at alternative options to meet their demand. India is a possible option, believes Kolaczowski, who has earlier worked on energy policy at the Polish Prime Minister’s Office.

However, he explains that the scale of capacity growth and invest-

preneurial scene that are supported by increasingly comprehensive policies, is well positioned [to become a major player]. The capital is available for good business solutions in the sectors,” he adds.

Appreciating the various central-level policies and regulations over the last few years to promote domestic manufacturing of advanced energy technologies and components, he says India has quite a comprehensive strategy. “The country promotes domestic mining by identifying 30 critical minerals and auctioning 20 blocks in 2023, with plans for 20 more in 2024. The government also

ute these benefits and enable viable business cases for investments,” explains Kolaczowski.

The key, he says, is a government mandate linked to demand certainty. Citing the example of the steel and cement sector, where he says 30% of goods is purchased through the public procurement process, there could be a mandate for clean products.

Kolaczowski’s assessment about India’s progress remains encouraging for the country to emerge as a global powerhouse in advanced energy solutions. **BT**

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